# ML1 Tips and Tricks

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## 1 MC Questions

There are several types of questions that tend to pop up in every exam, but in different shapes and forms. These are a couple of trends I have noted:

- Which model is suited to this task? Tests whether you understand which models are/can be used for regression, which for classification.
- Which models are (un)supervised?
- Which model can separate this dataset? This generally tests your knowledge of which models have linear decision boundaries and which have non-linear decision boundaries.
- When training model X, are we guaranteed to find a global optimum? This question tests whether you understand how a model is trained. Is there stochasticity involved? And if so, does that mean we reach a different solution every time? Is there a closed-form solution available to us, like in linear regression? And by the way, is that still true for logistic regression?
- Have you developed good intuition for overfitting/bias-variance tradeoff and can you tell me which of these models is overfitting?
- Which kernel function is suitable for this problem? How does it change as we vary the parameters?

## 2 Vector Calculus

- Three-step plan: (i) Component notation (ii) Differentiate scalarly (iii) infer shapes.
- The number of indices is conserved, except when summmed over. Also, the number of indices tells you what kind of object your derivative is.
- Always differentiate with respect to different indices and let Kronecker delta's do the work.

# 3 Likelihoods

In general, you will be able to factorise a likelihood. This makes optimisation much easier, since many terms will disappear when taking the log and the derivative. Some common ones include:

- $p(X) = \prod_N p(x_n)$ because i.i.d. assumption.
- $p(x) = \prod_D p(x_d)$ because of naive Bayes assumption.
- $p(x,t) = p(t)p(x|t) = \prod_K [p(t=k)p(x|t=k)]^{t_k}$. This is just a trick useful for multi-class classification.

# 4 Latent Variable Models

If there is either a sum or an integral in your likelihood, some variable is marginalised over. This means you're dealing with a latent variable model. These come in several forms, one of which is the mixture model.

If you ever see one of these in your likelihood, you should have a knee-jerk reaction: You should compute the posterior over the latent variable $p(z|x)$, which are called the responsibilities. You should then use this in the EM algorithm to iteratively optimise your parameters.

Also, if one of the parameters you're optimising over is a probability itself, don't forget your Lagrange multipliers.

# 5 SVMs

Solving constrained optimization ($\max_x f(x)$) problem via method of Lagrange multipliers for equality constraints ($g(x) = c$): It can be considered a three step process:

1. Identify the constraint. (Given in the question explicitly or need to be identified from the problem example: it is a probability and hence sums to 1)
2. Define Lagrangian $\mathcal{L}$:

   $$\mathcal{L} = f(x) + \mu_1(constraint1) + \mu_2(constraint2)\ldots$$

   Note: If there are $n$ constraints, then we will need $n$ Lagrangian multipliers.
3. Find stationary points of $\mathcal{L}(x, \mu_1, \mu_2 ...)$ Take derivative w.r.t $x$, set it to zero and find solution for $x$.

Constrained optimization with inequality constraint: This can be a thought of a six step process:

1. Identify the constraints. (Note: suppose we have two inequality constraints, $g(x) \geq 1$ and $h(x) \leq 0$, remember to swap signs when adding in the $\mathcal{L}$ term.
2. Define Primal Lagrangian $\mathcal{L}$

$$\mathcal{L} = f(x) + \mu_1(g(x) - 1) - \mu_2 h(x)$$

   where $\mu_1$ and $\mu_2$ are Lagrange multipliers for the constraints.
3. Identify KKT conditions;
   - $\mu_1 \geq 0, \mu_2 \geq 0$
   - $g(x) - 1 \geq 0, -h(x) \geq 0$
   - $\mu_1(g(x) - 1) = 0, -\mu_2 h(x) = 0$
4. Compute dual Lagrangian $\tilde{\mathcal{L}}(\mu)$, optimize w.r.t primal variable $x$ for a fixed dual variable $\mu$.
5. Solve dual problem

$$\mu^* = \underset{\mu}{argmin}\ \tilde{\mathcal{L}}(\mu) \quad \text{subject to} \quad \mu \geq 0$$

6. Maximize primal Lagrangian

$$x^* = \underset{x}{argmax}\ \mathcal{L}(x, \mu^*)$$

Note: If the problem changes from maximization to minimization the signs change. If the inequality is of the form $h(x) \leq 0$ the signs change as given above.

# 6 Principal Component Analysis

There are several things that are important. For example, if the covariance is expressed as

$$\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T,$$

then the columns of $\mathbf{U}$, namely $\mathbf{u}_k$, are orthogonal. Also remember that you're trying to maximise the variance left in the dataset. It's good to remind yourself of an ellipsoid, so that axes align with the variance.